

Active ants: dynamics

$$\begin{cases} \partial_t f = \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \sigma_\theta \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho + \tau \nabla^2 K * \rho v(\theta)) f) \\ f(t=0) = f_0 \end{cases}$$

- What happens as $t \rightarrow \infty$?

• Part 2 (Rakotomalala & W, in review, 2026):

What about existence of spots and lanes?

Bifurcation theory (e.g. Kielhöfer, 2004): Lyapunov-Schmidt method

O PDE $\Rightarrow$ ODE

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